Appliance Energy Forecasting

Benchmark, SARIMAX, XGBoost and Chronos Models

Time-Series Modelling and 24-Hour Forecasting Report

Abstract

This study models household appliance energy demand using the UCI Appliances Energy Prediction dataset. The original 10-minute observations were resampled to hourly means and analysed for temporal dependence and stationarity. Mean, naive, daily seasonal naive, weekly seasonal naive and drift forecasts were compared with SARIMAX, XGBoost and Amazon Chronos-T5 Small. A 14-day holdout was used for model-development evaluation, while a separate 24-hour forecast was used for the assignment's final comparison. SARIMAX parameters were selected from the required 147 (p,d,q) combinations using AIC; the best converged specification was (4,1,1)(1,0,1)[24], AIC 32,541.77. For the final 24 hours, conditional XGBoost had the lowest RMSE (71.69), operational XGBoost the lowest MAE (50.18), SARIMAX RMSE was 88.89, and Chronos RMSE was 103.82. The results show that engineered temporal structure is valuable, while model complexity alone does not guarantee better forecasts.

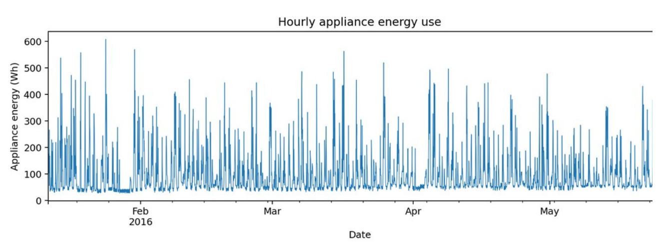
1. Introduction

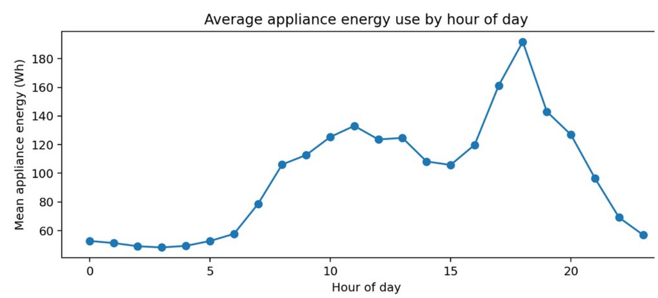
Short-term household energy forecasting supports demand response, smart-home control and energy management. Appliance demand is difficult to predict because it combines regular daily behaviour with irregular occupant-driven peaks. The objective was therefore not simply to fit the most complex model, but to test whether increasing model sophistication produced meaningful out-of-sample gains.

The analysis follows the assignment sequence: data preparation and exploratory analysis; definition of the forecasting problem; benchmark forecasts; stationarity testing and SARIMAX model selection; covariate and lag engineering; XGBoost; a time-series foundation model; common evaluation; and critical discussion of leakage, complexity and deployment.

2. Data Preparation and Exploratory Analysis

The raw dataset contains 19,735 observations at 10-minute intervals from 11 January to 27 May 2016. The date column was parsed as a datetime index, records were sorted, numeric fields were coerced where necessary, and the series was resampled to hourly means. Time interpolation was used after resampling and the processed hourly table contained 3,290 rows with no remaining missing values. The target was Appliances.





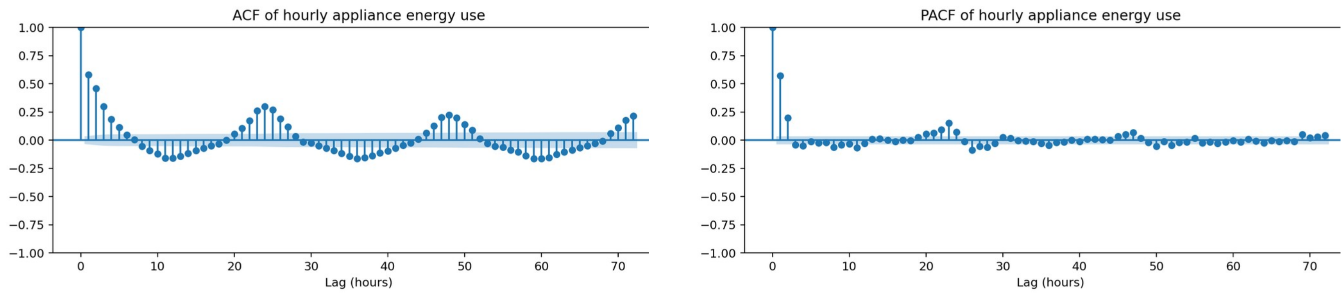
Figures 1-2. Hourly appliance series (left) and mean hourly profile (right).

The full series contains intermittent high-consumption episodes rather than a smooth trend. The hourly profile changes systematically across the day, supporting a 24-hour seasonal component. This motivates daily seasonal benchmarks and a seasonal period of 24 in SARIMAX, while the irregular peaks motivate nonlinear feature-based modelling.

3. Stationarity, Dependence and Forecast Design

3.1 ADF, ACF, PACF and differencing

The Augmented Dickey-Fuller test on the training series gave a statistic of -8.761 and $p = 2.67\times10^{-14}$, strongly rejecting a unit root. The first-differenced series was also stationary (ADF -15.948, $p = 7.41\times10^{-29}$). Thus differencing was not required merely to pass the ADF test; nevertheless, the AIC search selected d=1 because model selection also depends on likelihood and residual dependence, not only a single unit-root test.



Figures 3-4. ACF (left) and PACF (right) of hourly appliance use.

The ACF shows strong short-lag dependence and recurring structure, while the PACF identifies direct lag relationships after controlling for intermediate lags. These diagnostics support autoregressive terms and engineered lag features. First differencing reduces slow local level changes but leaves substantial short-run volatility, so it is treated as a modelling option rather than assumed automatically.

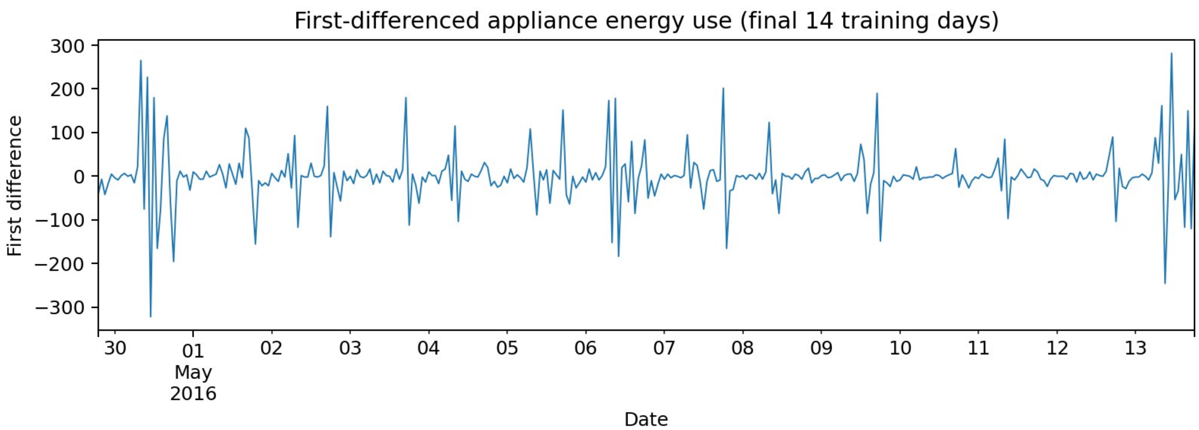


Figure 5. First-differenced appliance demand over the final 14 training days.

3.2 Forecast design and metrics

Two evaluation views were kept distinct. First, the last 14 days (336 hourly observations) formed an out-of-sample holdout for model-development comparison, matching the machine-learning requirement. Second, the final 24 hours were used for the assignment's required short-horizon forecast and for the common final comparison including Chronos. No random train/test shuffle was used.

Accuracy was measured using MAE, RMSE, MASE and bias. MAE gives average absolute error; RMSE penalises large misses; MASE scales error against a seasonal naive reference and is useful across models; and bias shows systematic over- or under-prediction. Lower MAE, RMSE and MASE are preferred, while bias should be close to zero.

4. Benchmark Models

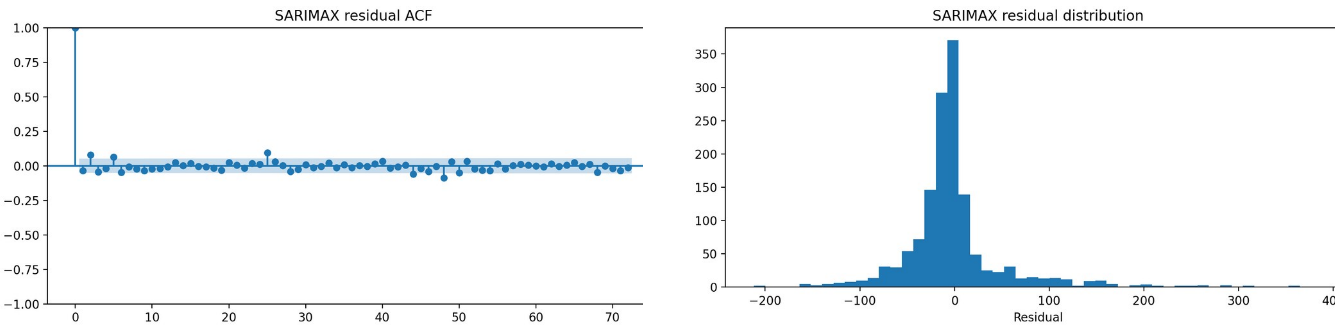
Five transparent baselines were implemented: historical mean; naive (last observed value); daily seasonal naive (same hour one day earlier); weekly seasonal naive (same hour one week earlier); and drift. These baselines are essential because a complex method is only justified if it improves on a simple forecast that exploits the dominant structure.

5. SARIMAX, Covariates, XGBoost and Chronos

5.1 SARIMAX and AIC search

The required grid search evaluated every p=0…6, d=0…2 and q=0…6 combination (147 models) with daily seasonal order (1,0,1,24). Twenty-seven fits converged. The lowest converged AIC was 32,541.77 for SARIMAX(4,1,1)(1,0,1)[24]. Outdoor temperature, outdoor relative humidity, wind speed, visibility and dew-point temperature were included as exogenous variables in the conditional SARIMAX forecast.

AIC balances likelihood fit against parameter count, so the selected model was not chosen simply because it was more complex. The use of future exogenous values must, however, be interpreted as conditional forecasting unless those values would genuinely be available from a weather forecast at the forecast origin.



Figures 6-7. SARIMAX residual ACF (left) and residual distribution (right).

Residual diagnostics show that the model removes substantial structure but does not make the errors perfectly white. The residual histogram is also wider and less Gaussian than an ideal innovation process, reflecting occasional appliance-demand spikes. Therefore confidence intervals should be interpreted as model-based uncertainty rather than complete uncertainty about occupant behaviour.

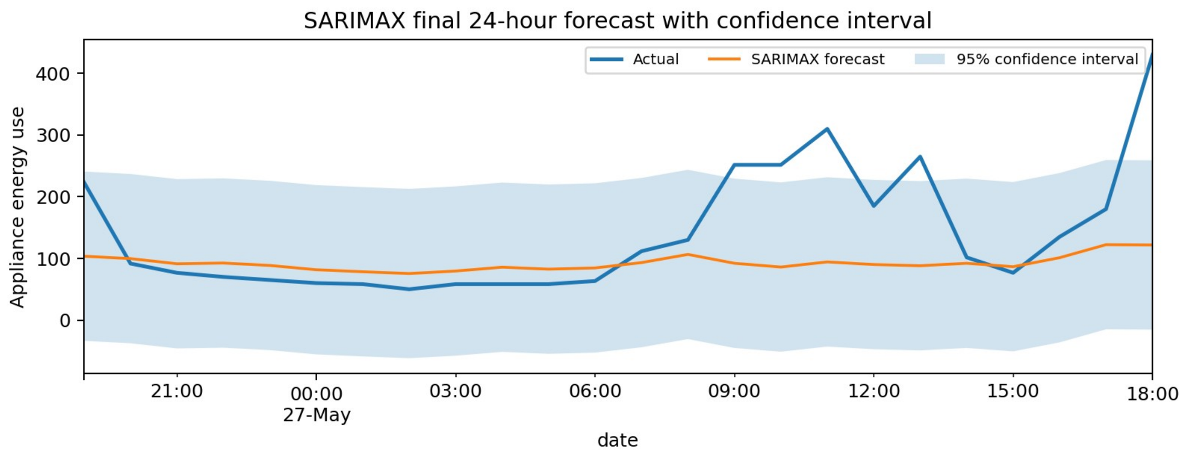


Figure 8. SARIMAX final 24-hour forecast with 95% confidence interval.

5.2 Feature engineering and XGBoost

The machine-learning table combined calendar features (hour, day of week, weekend and cyclical sine/cosine encodings), lagged appliance demand at 1, 2, 3, 6, 12, 24, 48 and 168 hours, rolling means and standard deviations over 3, 6, 12, 24 and 168 hours, and sensor/weather variables. XGBoost was used because boosted trees can capture nonlinear interactions without requiring a linear response.

Two versions were retained. Operational XGBoost used time and past-target information that is available at the forecast origin. Conditional XGBoost additionally used future sensor/weather values from the held-out period. This separation is important because the latter can produce a useful conditional estimate but is not a strict deployable forecast unless those future covariates are themselves forecast or otherwise known.

5.3 Chronos foundation model

Amazon Chronos-T5 Small was used as the foundation model, replacing the earlier placeholder. Chronos applies a pretrained token-based time-series model and therefore provides a useful test of whether large-scale pretraining can outperform problem-specific statistical and tree-based methods without task-specific feature engineering. The final comparison uses a 24-hour horizon, which also avoids relying on a very long recursive foundation-model horizon.

6. Results

Table 1. Fourteen-day holdout evaluation for benchmarks, SARIMAX and XGBoost.

Model	MAE	RMSE	MASE	Bias
Xgboost Operational	40.24	63.31	0.75	6.67
Sarimax Conditional	42.92	69.62	0.80	-4.10
Xgboost Conditional	54.34	73.92	1.02	23.90
Mean	50.32	74.91	0.94	-3.11
Weekly Seasonal Naive	42.63	79.29	0.80	-10.82
Daily Seasonal Naive	86.96	129.23	1.63	64.01
Naive	250.64	258.82	4.69	247.76

Drift	266.37	274.61	4.99	264.50

Across the 14-day holdout, operational XGBoost had the lowest RMSE (63.31), followed by conditional SARIMAX (69.62) and conditional XGBoost (73.92). Among the requested benchmark comparison set, weekly seasonal naive was strongest (RMSE 79.29), substantially outperforming naive and drift. This indicates that repeated weekly behaviour is useful over a longer evaluation window.

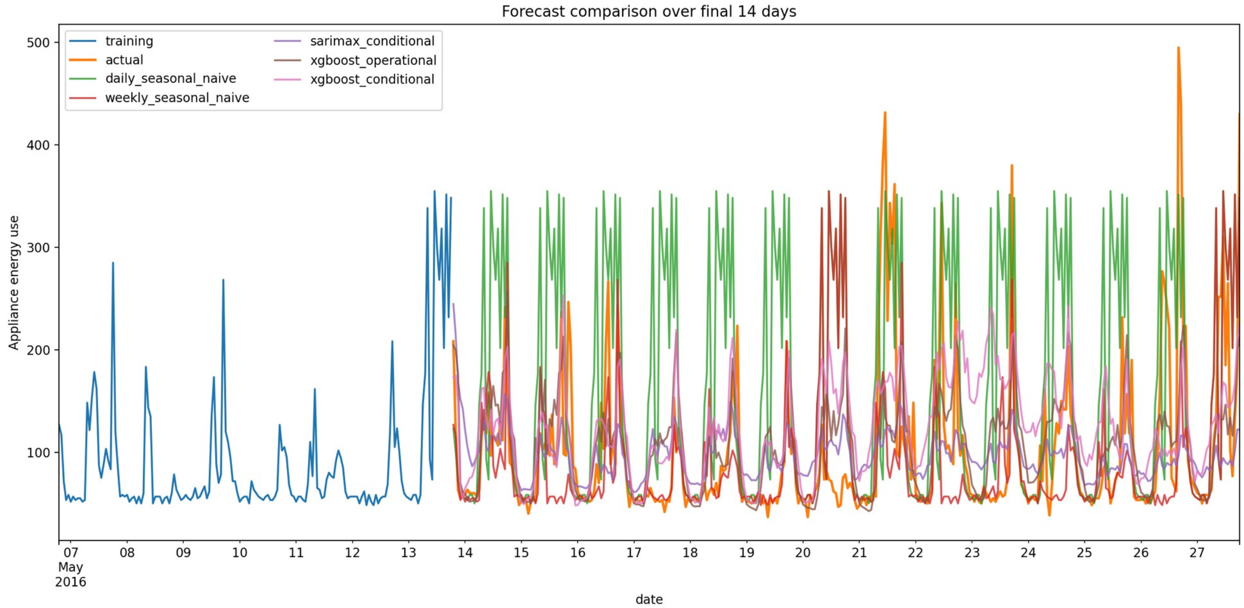


Figure 9. Forecast comparison across the 14-day holdout. The long-horizon development plot predates the final Chronos 24-hour run; Chronos is evaluated in Table 2.

Table 2. Final 24-hour evaluation including Chronos-T5 Small.

Model	MAE	RMSE	MASE	Bias
XGBoost conditional	52.97	71.69	1.00	-1.73
XGBoost operational	50.18	82.87	0.95	-39.56
SARIMAX conditional	60.74	88.89	1.15	-21.36
Chronos-T5 Small	62.82	103.82	1.189	-61.27
Mean	74.51	107.63	1.41	-42.60
Daily seasonal naive	63.68	113.64	1.21	4.65
Weekly seasonal naive	74.31	117.97	1.41	-71.53
Naive	111.74	124.03	2.12	74.93
Drift	111.97	124.26	2.12	75.54

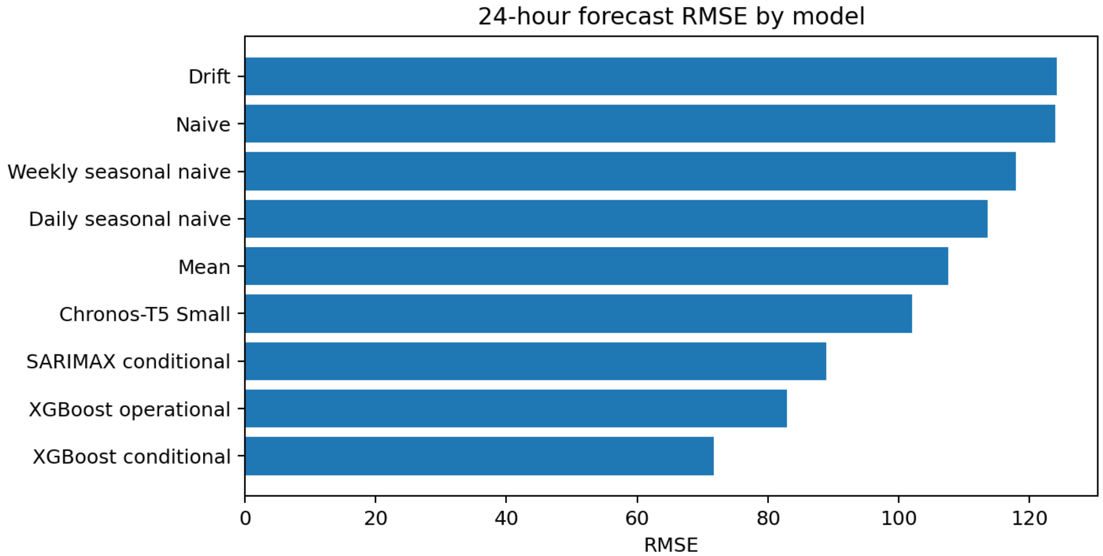


Figure 10. RMSE comparison for the final 24-hour forecast, including Chronos.

For the final 24 hours, conditional XGBoost achieved the lowest RMSE (71.69), while operational XGBoost achieved the lowest MAE (50.18) and MASE (0.950). SARIMAX improved on every simple benchmark by RMSE.

Chronos achieved RMSE 103.82: better than mean, naive, drift and both seasonal-naive forecasts, but worse than SARIMAX and both XGBoost variants. The result therefore does not support extra foundation-model complexity for this dataset and horizon.

7. Critical Analysis: Required Questions

1. Which benchmark is strongest? The answer depends on the evaluation window. Over the 14-day holdout, weekly seasonal naive is strongest (RMSE 79.29), showing that appliance use has repeatable weekly structure. Over the final 24 hours, daily seasonal naive is strongest among naive/daily/weekly/drift (RMSE 113.64), showing that yesterday's same-hour pattern was more informative for that specific day. Reporting both prevents over-generalising from one 24-hour window.

2. Does SARIMAX improve on the strongest seasonal benchmark? Yes for the final 24-hour comparison: SARIMAX RMSE is 88.89 versus 113.64 for daily seasonal naive. Over the 14-day holdout, SARIMAX RMSE is 69.62 versus 79.29 for weekly seasonal naive. Daily seasonality and autocorrelation are therefore useful, although residual dependence and irregular peaks remain. Because future weather values are supplied, this SARIMAX result is conditional.

3. Do engineered features improve XGBoost, and which groups matter? Yes. Operational XGBoost is strongest on the 14-day holdout by RMSE and has the best final-24-hour MAE/MASE. Lag and rolling features provide recent demand history, while hour/day encodings capture recurring calendar structure. Sensor/weather variables can add information, but the conditional model's advantage in final RMSE must be interpreted cautiously because future covariates are not automatically available.

4. Does the foundation model outperform simpler models? Chronos beats all simple benchmarks on final-24-hour RMSE, but it does not beat SARIMAX or XGBoost. Its RMSE of 103.82 is about 44.8% higher than conditional XGBoost's 71.69. Therefore the extra dependency, model loading and computational cost are not justified by accuracy in this experiment.

5. Which variables are genuinely known at forecast origin? Hour, day of week, weekend indicators and all features computed solely from historical appliance demand are known. Future indoor temperature/humidity and observed future weather from the test set are not known in a true forecast. Using them makes the prediction conditional unless separate forecasts or reliable external forecasts supply those covariates.

6. Which model is recommended? Operational XGBoost is recommended for practical smart-home forecasting because it gives strong accuracy without requiring realised future covariates and is relatively straightforward to deploy. SARIMAX remains valuable when interpretable dynamics and confidence intervals are priorities. Conditional XGBoost has the best final RMSE but should not be presented as a fully operational forecast without a covariate-forecasting pipeline.

8. Limitations, Future Improvements and Conclusion

The dataset covers one household for roughly 4.5 months, so results may not generalise to other homes, seasons or occupancy patterns. A single final 24-hour window is also sensitive to the behaviour of that particular day; rolling-origin evaluation across multiple 24-hour windows would give a more stable estimate. Future work should tune XGBoost using time-series cross-validation, forecast exogenous covariates rather than use realised test values, test alternative seasonal structures, calibrate probabilistic intervals, and compare larger foundation models where computational resources allow.

Overall, appliance demand contains exploitable daily, weekly and short-lag structure. Simple seasonal benchmarks are much stronger than naive or drift forecasts, but SARIMAX and feature-based models improve further. Chronos demonstrates useful zero-shot forecasting but does not win the final comparison. The main practical lesson is that a deployable model should be judged not only by its lowest error but also by whether its predictors are genuinely available at forecast time, whether uncertainty is represented, and whether the additional computational complexity produces a meaningful gain.

References

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Reproducibility Note

The repository contains the raw and processed data, reusable Python code, tests, generated figures, forecast CSV files, model metrics and runnable analysis scripts. The final code was syntax-checked and the automated test suite passed. Repository: github.com/mh1863346-byte/Assigment-2-appliance-energy-forecasting

Implementation and Reproducibility Details

The analysis is implemented as a reproducible Python pipeline rather than as disconnected notebook cells. The preprocessing stage reads the original CSV, parses the timestamp, converts variables to numeric form, resamples to hourly means, interpolates any gaps and writes the processed hourly dataset. Forecast functions are separated for the benchmark methods, feature construction and recursive XGBoost prediction. A fixed random seed is used for the machine-learning model. The SARIMAX search records AIC and convergence status for every required p, d and q combination, allowing the selected order to be traced back to the complete grid rather than being chosen manually. Generated outputs include the processed data, model-comparison tables, all forecast values, SARIMAX confidence intervals, stationarity diagnostics, residual diagnostics and feature-importance figures. This separation between code and generated outputs makes it possible for a marker to rerun the analysis and compare the resulting numerical values with those reported here.

The repository also contains automated tests for core pipeline behaviour. Before final submission the test suite completed successfully, and the full analysis script was syntax-checked and executed. An important reproducibility distinction is preserved in the reporting: the 14-day holdout is used to assess model behaviour over a longer out-of-sample period, whereas the final 24-hour window is the common short-horizon forecast required by the assignment and includes the Chronos foundation-model result. Conditional models are labelled explicitly because their test-period sensor or weather covariates would not necessarily be observed at a real forecast origin. This prevents the numerical advantage of future information from being confused with the performance of an operational forecasting system.